

Dynamics of Flow-Excited Hydraulic Gates: Loading and Identification

دینامیک دریچه‌های هیدرولیکی تحت تحریک جریان: بارگذاری و شناسایی

A proposed pair of independent but complementary doctoral concepts

Candidates: **Zina · Jamal** Proposed supervisor: **Prof. Reza Tarinejad** October 2026

Status

Proposed research direction for preliminary discussion, subject to supervisor confirmation. Two points remain open: (1) a short recent-literature confirmation review of both research gaps; (2) confirmation that neither scope overlaps with current students' work.

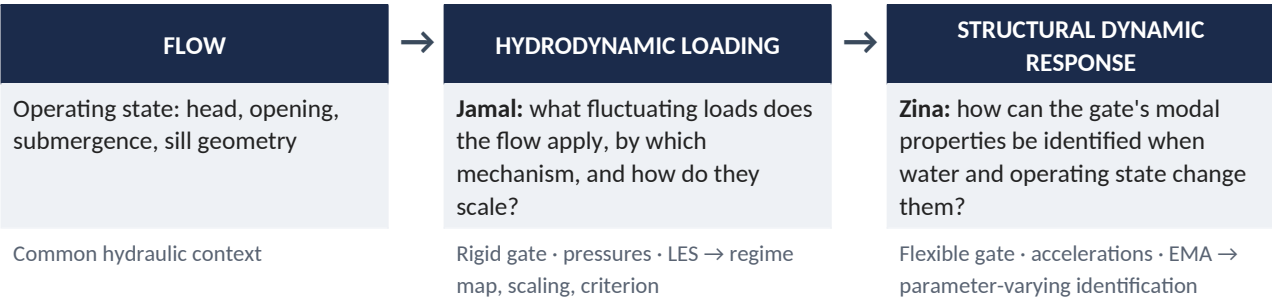
1. Why this programme

Prof. Tarinejad's publication record contains two active lines that are not yet connected:

- **Output-only identification of dam dynamics**, most recently transmissibility-based methods [1, 2, 4] and automated SSI [3].
- **Hydraulic control structures**: gate and weir discharge performance [5, 6] and outlet CFD verification [7].

The hydraulic work characterises time-averaged performance; the identification work addresses dam bodies under seismic or ambient input. Flow-excited gates sit between the two: they are excited by fluctuating flow, and their dynamic properties change with water level and operation.

2. Scientific logic



Each thesis addresses **one link** of the chain: Jamal characterises the excitation, Zina identifies the structure. Neither thesis needs the other's results. Zina uses synthetic loads defined independently, and Jamal uses a rigid gate whose dynamics play no role in his question.

3. The two doctoral concepts

	ZINA	JAMAL
Discussion title	Operational Modal Identification of Hydraulic Gates under Varying Flow Conditions	Unsteady Hydrodynamic Loading of Sluice Gates with Bottom Sills
Research gap	Transmissibility OMA assumes the structure is unchanged across loading conditions; in gates, changing the operating state changes the added mass and damping as well	A regime-based, dimensionless description of fluctuating loads on sill-equipped gates, and an efficiency-load criterion, appear not to be available
Core question	How can structural poles be separated from operating-state pole shifts and flow-excitation peaks?	How do sill geometry and submergence change load magnitude, frequency and mechanism, and where is the efficiency-load optimum?

	ZINA	JAMAL
Working hypothesis	Modelling poles as smooth functions of a measured hydraulic state θ gives unbiased identification across operating states	A sill shifts separation control from the gate lip to the sill crest; frequencies follow regime-specific Strouhal scaling; a favourable trade-off region exists
Expected contribution	Parameter-varying transmissibility identification for water-interacting structures	Regime map, scaling relationship and efficiency-dynamic-load criterion

4. Scope separation

Component	Zina	Jamal	Shared
Research gap	Identification of time-varying water-interacting structures	Mechanism and scaling of fluctuating loads	—
Scientific question	Separating structural poles from operating-state effects	Load regimes, scaling and the trade-off	—
Hypothesis	Parameter-varying pole model $\lambda r(\theta)$	Separation-control shift; regime Strouhal scaling	—
Test specimen	Flexible instrumented gate	Rigid gate with interchangeable sills	Gate family / outer geometry
Sensors	Accelerometers; impact hammer	Pressure transducers	DAQ hardware
Measurements	Accelerations; EMA FRFs	Wall pressures; discharge; depths	Hydraulic-state labels
Numerical modelling	Structural FE with added mass; synthetic loads	LES/DES with VOF (CFD)	—
Analysis	System identification algorithms	Spectral load analysis; dimensional analysis	—
Validation	FE exact poles; EMA ground truth; public dam records	Measured spectra; held-out configurations	—
Expected results	Identification method and protocol	Regime map, scaling law, criterion	—
Journal papers	Z1–Z3 (own data, first author)	J1–J3 (own data, first author)	Optional joint paper only

Independent executability

If Zina's work is delayed by a year, can Jamal still finish?	Yes. His experiments, CFD and load model use his own rigid specimen and pressure data, and need no modal information.
If Jamal's work is delayed by a year, can Zina still finish?	Yes. Year 1 is theory and FE; her laboratory runs need only flume flow and her own specimen; Paper Z3 uses public dam records; her loads are synthetic.

Each thesis is designed to remain independently executable, with clear scope separation and independent data ownership.

5. Shared resources

- **Flume and water supply**, with shared scheduling. Runs may be simultaneous, but each candidate records their own quantities.
- **General gate family and initial drawings**, with the same outer dimensions but **separate specimens**: flexible for Zina, rigid with sill inserts for Jamal.
- **DAQ infrastructure and general laboratory setup**, with separate channels and files.
- **Common baseline hydraulic conditions** (head, opening, submergence, discharge), logged once and used by both only as labels.

No measured response, load, model or result is shared in a way that would create dependency. One optional joint paper (for example, a resonance-risk discussion combining load spectra with identified frequencies) could follow later, but it would not form part of either thesis's core contribution.

6. Minimum laboratory requirements

Item	Zina	Jamal
Flume	ESSENTIAL: hold still water at 4–6 levels; run 4–6 flow states	ESSENTIAL: tailgate for free/submerged control; adequate discharge range
Gate model	ESSENTIAL: one flexible plate gate (light, measurable modes)	ESSENTIAL: one rigid gate; sill inserts (3 heights × 2 lengths)
Response / load sensors	ESSENTIAL: 6–8 accelerometers; impact hammer with force sensor. USEFUL: strain gauges. OPTIONAL: laser vibrometer	ESSENTIAL: 6–8 flush pressure transducers with adequate bandwidth. USEFUL: two-component load cell
Hydraulic measurements	USEFUL: head and discharge as θ labels	ESSENTIAL: discharge meter; water-level gauges. USEFUL: high-speed camera. OPTIONAL: ADV/PIV
DAQ	ESSENTIAL: synchronised, ≥ 8 channels	ESSENTIAL: synchronised, ≥ 8 channels (shared hardware possible)
Operating-state variation	ESSENTIAL: water level and opening at 4–6 states	ESSENTIAL: opening (3) × flow state (2–3)
Minimum configurations	About 4–6 still-water EMA levels + 4–6 flow states (about 20–30 runs)	7 geometries × 3 openings × 2–3 flow states (about 40–60 runs)
Computation	Workstation (FE, identification)	Workstation or cluster for 6–8 LES/DES cases

The experimental scope is deliberately bounded. Instrumentation requirements will be refined after confirmation of existing laboratory resources.

7. Publication architecture

	Zina	Jamal
Paper 1	Parameter-varying transmissibility formulation and numerical benchmark	Experimental fluctuating loads and regime identification
Paper 2	Laboratory validation with EMA ground truth	Separation-control mechanism and Strouhal scaling (experiment + LES)
Paper 3	Dam application with reservoir-level-dependent records	Efficiency–dynamic-load criterion; prototype-scale interpretation

8. Open points and alternatives

- Recent-literature confirmation** of both gaps through a short structured review. If the plain-sluice framing of Jamal's concept proves already covered, the same question would be studied for labyrinth sliding gates with sills [5].
- Scope:** confirmation that neither topic overlaps with current students' work (for example, in transmissibility identification or gate hydraulics).
- Facilities:** flume, pressure transducers, accelerometers and DAQ, available or accessible.

Alternative if laboratory access is not available: both concepts would become identification-based and require no pressure campaign. Zina would study uncertainty-quantified transmissibility identification from seismic dam records; Jamal would study the gate-identification concept, validated numerically and on an accessible regulator gate. If only Zina's topic overlaps with a current student's work, Zina would move to the seismic-uncertainty concept and Jamal's concept would remain unchanged.

Key supervisor publications referred to

- [1] Damadipour M., Tarinejad R., Aminfar M.H. (2021). Seismic modal identification using a new approach based on weighted transmissibility. *Earthquake Engineering & Structural Dynamics*, 50(7). <https://doi.org/10.1002/eqe.3435>
- [2] Damadipour M., Tarinejad R., Aminfar M.H. (2022). Structural system identification based on combining weighted transmissibility and wavelet transform. *Structural Control and Health Monitoring*, 29(2), e2868. <https://doi.org/10.1002/stc.2868>
- [3] Tarinejad R., Hafezi A., Pourgholi M. (2025). Automated modal analysis of an arch dam using a hybrid SSI framework based on error analysis and fuzzy clustering. *Journal of Vibration and Control*.
- [4] Tarinejad R., Amanzad F. (2025). Identification of operational modal parameters using multi-reference weighted Hermitian transmissibility functions. *Journal of Sound and Vibration*, 615, 119212. <https://doi.org/10.1016/j.jsv.2025.119212>
- [5] Abbaszadeh H., Tarinejad R. (2025). Soft computing and predicting labyrinth gates discharge coefficient. *Water Resources*, 52(4), 701–714. <https://doi.org/10.1134/S0097807824604734>
- [6] Abbaszadeh H., Tarinejad R., Roushangar K., Daneshfaraz R. (2026). Investigation of the effects of internal structure and material properties on the hydraulic performance of rectangular gabion weirs: experimental and numerical study. *Water Resources Management*, 40, 376. <https://doi.org/10.1007/s11269-026-04667-3>
- [7] Vaseti M.M., Akhtari A.A., Tarinejad R. (2026). Optimized mesh zoning for dam bottom outlet CFD simulations using Richardson extrapolation and grid convergence analysis. *Engineering Reports*, 8(6). <https://doi.org/10.1002/eng2.70800>